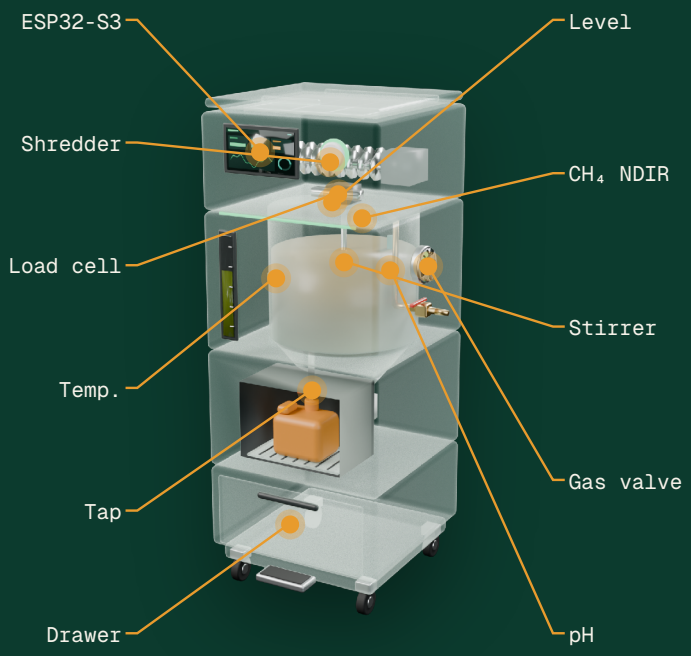
03 Inside ARUNA

Food waste is shredded, then digested by bacteria without oxygen at 30–38 °C. Sensors keep the process stable.



01 pH below 6.5? Feeding pauses automatically.

02 Gas leak > 5,000 ppm? Valve closes.

06 Specifications

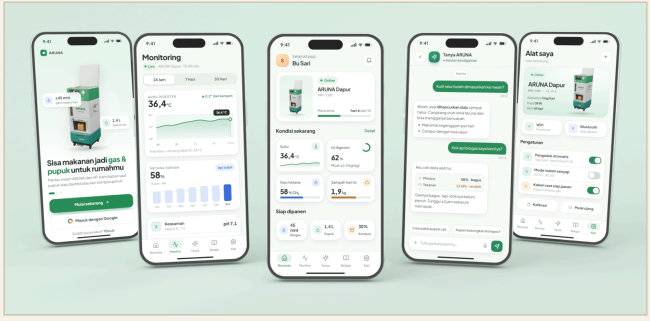
Size	44 × 44 × 112 cm
Capacity	1.5 kg/day (peak 3)
Working volume	100 L
Retention time	±34 days
Sensors	T · pH · CH_4 · level · mass
Controller	ESP32-S3 · Wi-Fi · BLE

SUPPORTS SDGS 7.1 · 11.6 · 12.3 · 13.2

7 11 12 13

TRY THE APP

aruna-id.web.app



ISIF International Science and Invention Fair

IYSA Indonesian Young Scientist Association

*Design prediction at 1.5 kg/day. 3D character: sthaarpit, Sketchfab, CC-BY 4.0.

ARUNA

Leftovers, *made energy.*

A smart home biodigester: cooking gas, liquid fertilizer and compost from your own kitchen.



BIOGAS · FERTILIZER · COMPOST

01 The problem

53 kg

of food is thrown away per person each year in Indonesian homes.

UNEP FOOD WASTE INDEX 2024

14.7 Mt household food waste / year	**≈ 40 %** of municipal waste is food
1.7 Gt CO_2e from food loss & waste	**CH_4** released as it rots in landfill

In ARUNA, that methane is captured as a cooking flame.

02 One machine, three outputs

From 1.5 kg of food waste a day, the design calculation predicts:

214 L biogas per day	**43 min** cooking per day
3.0 kg LPG saved / month	**0.72 t** CO_2e avoided / year



Biogas — kitchen stove

Liquid fertilizer — dilute 1 : 10

Compost — bottom drawer

04 How to use



01
Feed
Open the lid, add leftovers.



02
Digest
Shredded & stirred, ±34 days.



03
Monitor
Status & alerts on your phone.



04
Fertilize
Dilute 1 : 10, water plants.



05
Cook
Blue flame, ±43 min/day*.

Two minutes a day. The ARUNA app tells you when fertilizer and compost are ready.